RP0 — The hypotheses of Revision_Deflation, the assumptions added to them, what was repeated, and what is open

Provenance of PROJECT spacetimes, Revision_Deflation: the map of the revision, and the explanation required by Hard Rule 7 for the one item that was not carried out

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1. The hypotheses

On November 12, 2026 the author paused all stages of the main project (they are recorded in the provenance P11 of the main project and are to be completed later) and instructed that every calculation of the main task be repeated, in the new folder `Revision_Deflation`, under new hypotheses. In the assistant's words, with the author's formulas:

- On the 4+4 spacetime and on the 7+1 spacetime alike, the coordinates $x5$, $x6$ and $x7$ **deflate exponentially**.
- Minkowski space is replaced by a pseudo-Riemannian space with a **diagonal metric** whose three components along these coordinates are, for the 4+4 spacetime,

$$g_{55} = -\text{const}^2\, e^{-2\,\text{g55}\,x4}, \qquad g_{66} = -\text{const}^2\, e^{-2\,\text{g66}\,x4}, \qquad g_{77} = -\text{const}^2\, e^{-2\,\text{g77}\,x4},$$

  with $\text{g55}, \text{g66}, \text{g77} > 0$ and constants that can be different. (The author first wrote the three components with a positive sign and then corrected the sign for the 4+4 spacetime.)
- Everything that the main task demands is to be mirrored, including two PRD manuscripts with the titles "16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Deflating Spacetime" and "… on 7+1 Deflating Spacetime".

How the hypotheses were read.

1. *The names.* g55, g66, g77 are the author's names for the three rates, and they are used exactly so, upright and without subscript. The components of the metric are written g_{55}, g_{66}, g_{77}. (The assistant at first replaced the author's names by a Greek letter, which the author rejected; error R-E4 of the provenance RP9.) The author's word const stands for three constants, which are named const5, const6, const7.
2. *The sign on 7+1.* The positive sign of the author's first formula is the sign of these components of the metric in the convention in which space-like components are positive. The main project writes the 7+1 metric with the time-like component positive, $\eta_{1671} = \text{diag}(-1,-1,-1,-1,+1,-1,-1,-1)$. In that convention the three components are negative on 7+1 as well. The geometry — a proper length along $x5$ that shrinks as $e^{-\text{g55}\,x4}$ — is the same in either convention.
3. *The other coordinates* keep their flat metric: $x0$ (the hidden space coordinate), $x1, x2, x3$ (space) and $x4$ (the ordinary time).
4. *"Fresh calculations".* Nothing was taken over from the main project without being computed again on the deflating spacetimes; where a result of the main project holds unchanged, the program of the revision verifies that it does (for example, all 106 checks of the Dirac matrices, the 1170 exchange amplitudes, the internal symmetry).

2. What the revision contains

document	content
`dirac matrices deflating` (DMD)	geometry of the deflating spacetimes; Dirac matrices, spin connection; the Dirac equations of dirac1644d and dirac1671d; their solutions; symmetries
`supplement/RS1`	the Vorton-Scalar and the superfluid: reduction to four dimensions, the deflation clock, pairing, gap, bound pair, order parameter, vortices, dynamics
`supplement/RS2`	the Higgs hypothesis H2 on the deflating spacetimes
`supplement/RS3`	cosmology: bounds on the rate, Hubble constant, dark matter
`supplement/RS4`	the effective action of Cornwall, Jackiw and Tomboulis along the clock
`provenance/RP0`	this document
`provenance/RP1`	MASTER provenance of the symbols of the revision (Hard Rule 5)
`provenance/RP2, RP3`	the PDF files and the notebooks used
`provenance/RP8`	the source code: what computes what, how to compile and run it
`provenance/RP9`	the error log of the revision
`manuscript/`	the two PRD manuscripts
`README`	where to start

Every document exists as `.md`, `.tex` and `.pdf` (Hard Rule 1).

3. The assumptions that were added, and the hypotheses that were tested

The author's hypotheses fix the metric. Three further assumptions were needed in the cosmological part, and one hypothesis of the main project was tested again. They are kept apart in every document.

label	statement	whose	where used
A1	ordinary space $x1, x2, x3$ expands with the scale factor of standard cosmology	assumption of the revision	RS3
A2	gravity is the Einstein theory in eight dimensions, so that Newton's constant is G_8 divided by the volume of the extra coordinates and grows at the rate gsum	assumption of the revision	RS3
HV	the vacuum value of the Vorton-Scalar sets the electroweak scale	assumption of the revision, examined only to show its consequence	RS3, Sec. 3
H2	the superfluid state is the Higgs field	the author's hypothesis (main project)	RS2; RS3, Sec. 3

4. What mirrors what

The program of the main project has eleven commands. The table lists what each computes and where the revision repeats it. The program of the revision is `deflation` (folder `code/deflation-rs`); its commands are `matrices`, `modes`, `history`, `quench`, `cosmology`, `symmetry`, `channels`, `matter`, `paired`.

command of the main project	what it computes	repeated in the revision by	status
`matrices`	the Dirac matrices and their 106 checks; pair classification	`deflation matrices`: the same 106 checks again, and 47 of 47 checks of the geometry, of the coordinate matrices, of the spin connection and of the Dirac operators	done (DMD)
`channels`	exchange amplitudes of the Vorton-Scalar in 15 representative pair channels on each spacetime	`deflation channels` (checks `def-K1` to `def-K3`)	done (RS1, Sec. 3)
`gap`	the gap equation	`deflation history` (checks `def-H2`, `def-H3`)	done (RS1, Sec. 3)
`dimer`	the bound pair	`deflation history` (`def-H1`, `def-H4`), `deflation matter` (`def-D3`)	done (RS1, Sec. 4)
`symmetry`	the internal symmetry of the sector	`deflation symmetry` (`def-S3`; new: `44d-S1`, `S2`, `71d-S1`, `S2`)	done (RS2; DMD, Part 5)
`cjt`	the effective action: paired state, matter, temperature, vacuum, test of the method	`deflation paired` (`def-P1` to `def-P7`), `deflation history` (`def-H5` to `def-H8`)	done (RS4)
`halos`	estimates for dark-matter halos; self-interaction	`deflation matter` (`def-D1`, `def-D2`)	done (RS3, Sec. 6)
`background`	the cosmological background: sound horizon, angular scale	`deflation cosmology` (`def-C1` to `def-C4`), independently checked	done (RS3, Secs. 1 to 4)
`superfluid`	the superfluid as a cosmological fluid	`deflation matter`: growth of structure with the Newton's constant of the moment (`def-D4`), table of drifts (`def-D5`)	done in part (RS3, Secs. 5, 6)
`spectrum`	spectra of the microwave background with the Einstein–Boltzmann program of the main project	—	**not done** (Sec. 5)
`forecast`	forecast of constraints from mock data	—	**not done** (Sec. 5)

New in the revision, without counterpart in the main project:

command	what it computes	where
`deflation matrices`	geometry: curvature, Einstein and Lanczos tensors, Killing vectors, geodesics	DMD, Part 1
`deflation modes`	exact and numerical solutions of the Dirac equations: creation of particles (7+1), extinction (4+4)	DMD, Part 4
`deflation channels`	modes of the scalar: static source, creation, bounded solutions	RS1, Sec. 2
`deflation symmetry`	which rotations and boosts survive; vortex zero modes; reduction of the couplings	DMD, Part 5; RS1, Secs. 1, 6
`deflation quench`	whether the paired state can follow the deflation	RS1, Sec. 7
`deflation cosmology`	bounds on the rate of deflation; the exponent of the electroweak scale under H2	RS3, Sec. 3; RS2, Sec. 5

In total the program of the revision performs 217 checks, and the independent Python script 33 of 33; all pass.

The documents of the main project and their counterparts:

main project	revision
`dirac matrices` (DM)	`dirac matrices deflating` (DMD)
supplements S1 to S4	supplements RS1 to RS4

main project	revision
the two manuscripts on the flat 4+4 and 7+1 spacetimes	the two manuscripts on the deflating 4+4 and 7+1 spacetimes (folder `manuscript`), built from one source (`docs_src/manuscript`) by `code/scripts/build_manuscripts.py`
provenances P0 to P3, P8, P9	provenances RP0 to RP3, RP8, RP9
provenance P11 (paused stages)	none: nothing of the revision is paused; the one open item is Sec. 5 below

5. The item that was not carried out, and why (Hard Rule 7)

What. The commands `spectrum` and `forecast` of the main project compute the temperature and polarization spectra of the microwave background with an Einstein–Boltzmann program written for the main project, and forecast from them how well the parameters of the model can be constrained. The revision has not repeated these two calculations on the deflating spacetimes.

Why the hypotheses do not determine it. The background calculation of RS3 uses one of Einstein's equations, the component along the time. The source that keeps the three rates of deflation constant enters it only through its energy density, which the adjusted cosmological term absorbs (RS3, Sec. 2), and not through its stresses. The perturbations of the metric obey the other components as well, and those contain the stresses of that source and their perturbations. The hypotheses of the revision prescribe the metric and say nothing about the source; assumption A2 does not supply it either; and if gravity is of the Einstein–Gauss–Bonnet kind, for which the deflating spacetime with equal rates needs no source at all (DMD, Sec. 1.4a), the equations of the perturbations are different again. A spectrum of the microwave background on the deflating spacetime can therefore be computed only under a further assumption about the source or about the theory of gravity, and that assumption is the author's to make. The simplest candidate would be: the scale factors of $x5$, $x6$, $x7$ are not perturbed, and the perturbations of ordinary spacetime obey the four-dimensional equations with the Newton's constant of the moment. Whether that is what the author intends is not known.

Why it cannot be done by changing parameters. Under assumption A2 Newton's constant depends on the time. In an Einstein–Boltzmann calculation the gravitational coupling enters the expansion rate and every source term of the metric perturbations, while the rates of Thomson scattering and of recombination, which depend on atomic physics, must *not* change. The program of the main project has one constant of gravitation built into its units. A correct repetition requires a copy of four of its modules (background, Boltzmann hierarchy, line-of-sight spectra, forecast) in which the gravitational coupling is a function of the scale factor, and a new validation of that copy against the unchanged program at zero rate.

Why it was not done. The main project is paused by the author's instruction, so its program was not to be changed; the copy would have to be made and validated inside the revision. Because the calculation needs an assumption that the hypotheses do not contain, the assistant has not chosen one and computed with it; it stops at this point and asks, as Hard Rule 7 prescribes.

What is established without it. The effect of the deflation on the *background* — the sound horizon, its angular scale and the inferred Hubble constant — is computed (RS3, Sec. 2) and confirmed by an independent program. At the largest rate that lunar laser ranging allows under A2, Newton's constant at last scattering was 0.9969 of its present value. The spectra would change, beyond the shift of the angular scale, by effects of that relative size.

What the author is asked to decide. (1) Whether the two calculations are to be carried out. (2) If so, under which assumption about the perturbations: the candidate named above, or another. Until then the item is open, and no document of the revision claims a result about the spectra.

Three smaller items are open as well and are stated where they arise: the simulation of the superfluid as a cosmological fluid was repeated only through the growth of structure (RS3, Sec. 5); the

exponent n_{EW} of RS2 is known in closed form only for the scale-invariant case; and the stability of the Einstein–Gauss–Bonnet solution of DMD, Sec. 1.4a, has not been examined.

6. How the revision reaches GitHub

Hard Rule 2 demands a push as soon as any object is updated. As explained in the provenance P0 of the main project, the assistant has no permission to push to `github.com/once-ere/spacetimes`; every commit is transferred to the author's Mac as a git bundle within about a minute and applied to the local repository there, and a loop that the author runs in his Terminal pushes it. The same procedure was used for every commit of the revision. The commits of the revision are listed by

```
cd "/Users/nsh/Developer/github/spacetimes"
git log --oneline -- Revision_Deflation
```